IT Workshop II (Cloud Computing): Cloud Concepts and Technologies

August 12, 2026

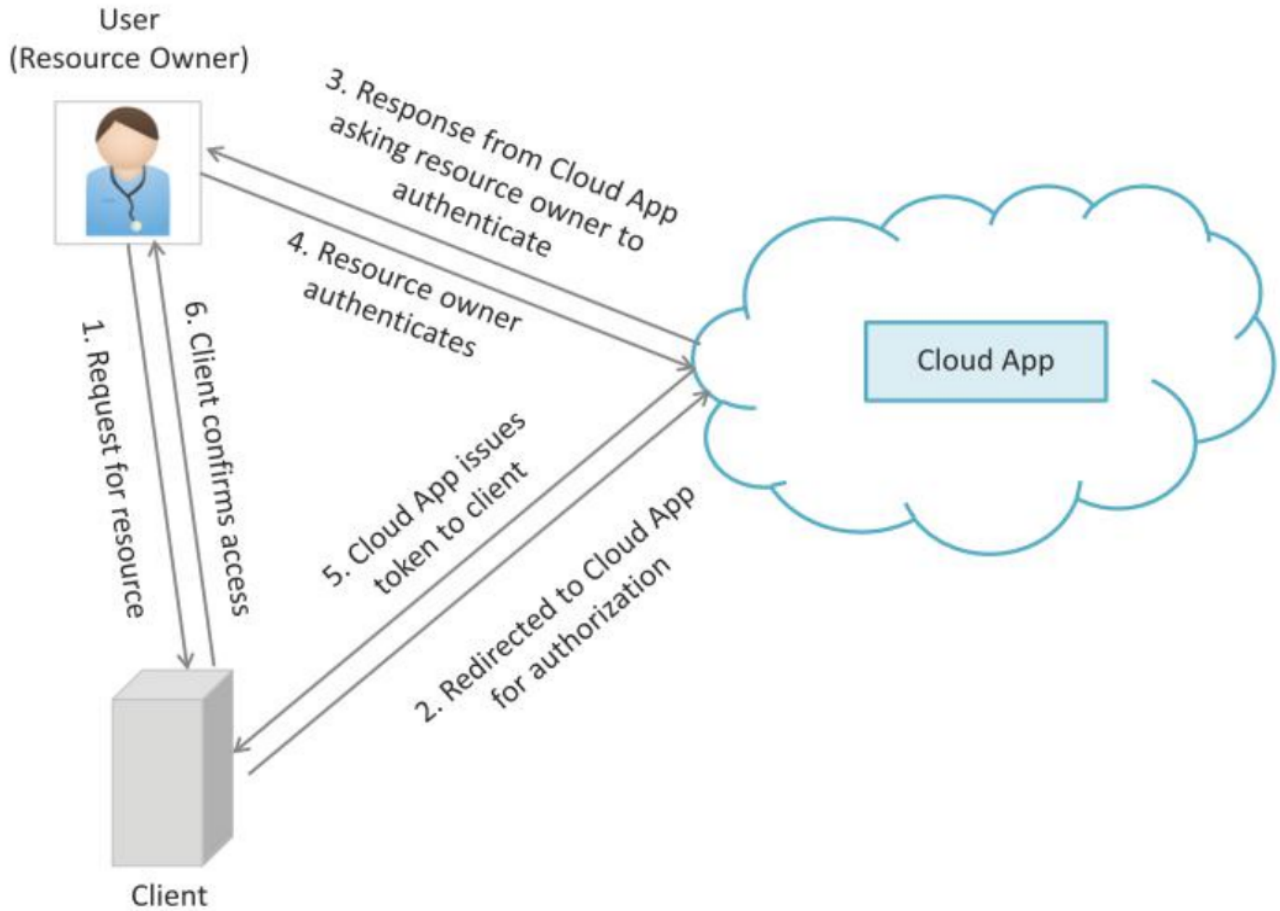
Identity and Access Management

- IDAM is enabled by a number of technologies such as OpenAuth, Role-based Access Control (RBAC), Digital Identities, Security Tokens etc.
- **OAuth:** It is an open standard for authorization that allows resource owners to share their private resources stored o one site with another site without handling out the credentials.
- In the OAuth model, an application (which is not the resource owner) requests access to resources controlled by the resource owner.
- The resource owner grants permission to access the resources in the form of a token and matching shared-secret.

Identity and Access Management

- Tokens make it unnecesary for the resource owner to share its credentials with the application.
- Tokens can be issued with a restricted scope and limited lifetime, and revoked independently.
- **RBAC:** It is an approach for restricting access to authorized users.
- A user who wants to access cloud resources is required to send his/her data to the system administrator who assigns permissions and access control policies which are stored in the *user roles* and *data access policies* database respectively.

Identity and Access Management



IAM User Access Key and AWS CLI

- Go to **Identity and Access Management (IAM)**. Select the user. Go to security credentials. Create an access key. Download the .csv file having the access key.
- Install **AWS CLI** on your local system.
- In a terminal in your local system, enter the following command:
  aws configure
  Here, give your access key details (read from the .csv file).

Python for Amazon Web Services

- Boto is a Python package that provides interfaces to Amazon Web Services (AWS).
- Boto supports many AWS services like Amazon EC2, Amazon EMR, Amazon RDS, AWS IAM, Amazon S3 etc.
- **Amazon EC2:** It is an Infrastructure as a Service (IaaS) provided by Amazon. EC2 delivers scalable, pay-as-you-go compute capacity in the cloud. The compute capacity is provided in the form of virtual machines that are launched in Amazon's cloud computing environment.

Python for Amazon Web Services

- We can install Boto using the following command.
  pip install boto3
- In *Identity and Access Management (IAM)*, we can create an access key for an IAM user. The .csv file having the access key can be downloaded.
- From the access key file, we need to note *AWS Access Key* and *AWS Secret Access Key*.

Python Program for Launching an EC2 Instance

```
import boto3

ec2 = boto3.client("ec2",
                   "ap-south-1",
                    aws_access_key_id="<enter access key id>",
                    aws_secret_access_key="<enter secret access
key>")

conn = ec2.run_instances(InstanceType="t3.micro",
                         MaxCount=1,
                         MinCount=1,
                         ImageId="ami-0f918f7e67a3323f0",
                         KeyName="<enter key name>",
                         SecurityGroups=[
                          "<enter security group name>"
                         ]
                        )

print(conn)
```

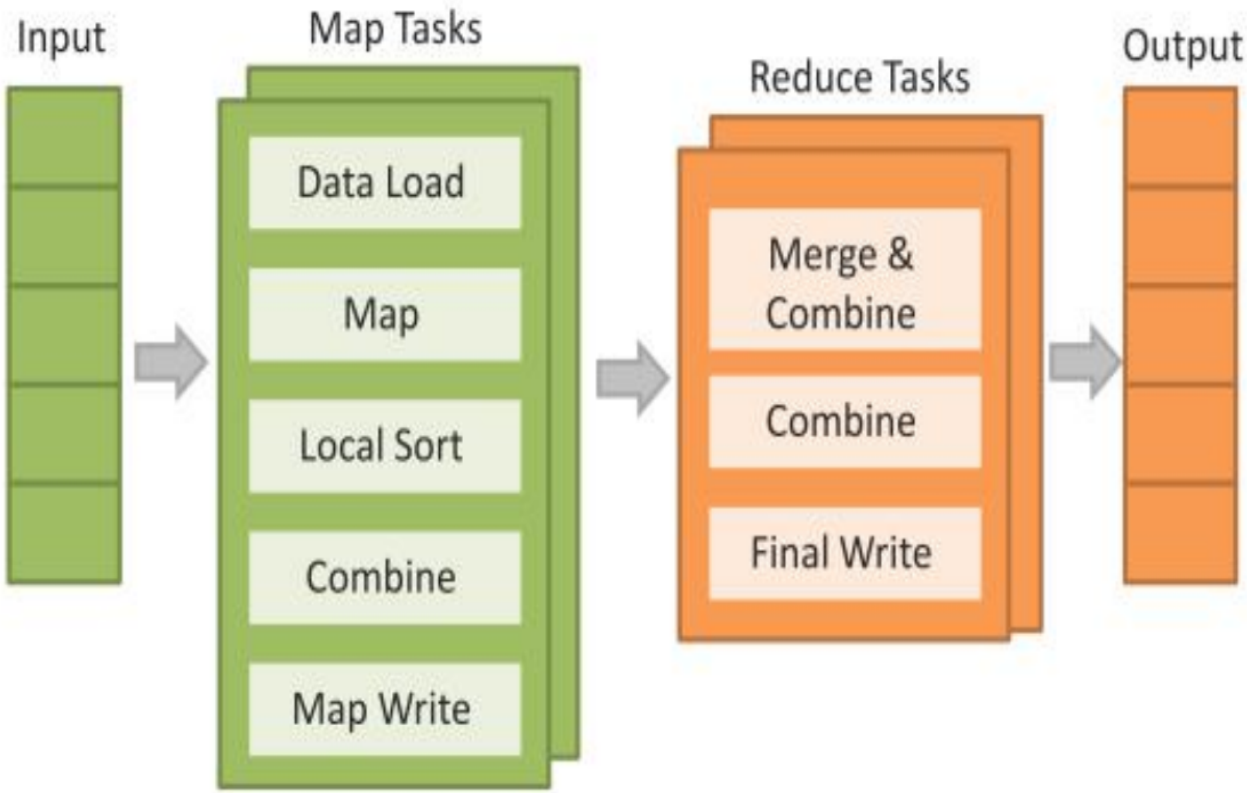
Map Reduce

- MapReduce is a parallel data processing model for processing and analysis of massive scale data.
- MapReduce model has two phases: Map and Reduce.
- The input data to the map and reduce phases is in the form of key-value pairs.
- In Map phase, data is read from a distributed file system, partitioned among a set of computing nodes in the cluster and sent to the nodes as a set of key-value pairs.

Map Reduce

- The Map tasks process the input records independently of each other and provide intermediate results as key-value pairs.
- The intermediate results are stored on the local disk of the node running the Map task.
- When all the Map tasks are completed, the Reduce phase begins in which the intermediate data with the same key is aggregated.

Map Reduce


Input
Map Tasks
Data Load
Map
Local Sort
Combine
Map Write
Reduce Tasks
Merge & Combine
Combine
Final Write
Output

The End